

Project Title: Smart Weather Monitoring System					
Phases	Task Name			Date	Status
Phase 1	Project Requirement				
		a.	Project abstract Preparation		
		b.	Setup Embedded Linux Development Environment		
Phase 2	Add LM35 Temperature sensor on target board				
	Step 1	Study Phase			
		a.	High Level Analysis		
			Understand Basics/terminology of ADC controller and prepare a document. Deliverable: document		
		b.	Low Level Analysis		
			Study ADC controller in AM3358 Technical Reference Manual and undestood the below things. ADC controller specifications ADC Functional Block Diagram ADC Controller Register Programming model Identify LM35 interface with ADC channel with the help of schematic diagram Prepare ADC-AM3358 Functional Block diagram Deliverable: document - ADC initalization - ADC Operation		
	Step 2	Implementation			
		a.	Bootloader stage		
			Enable ADC controller mux configuration and clock in u-boot source code.		
		b.	Kernel stage		
			DTS Modification: Add ADC device controller mux configuration if needed, base address, interrupt number, Chaneel number and low level device driver compatibility name in device tree source code. Deliverable: patch		
			Driver Modifications: Identify low level driver and find out low level driver comes which Linux subsystem. Deliverable: patch		
	Step 3	Application			
			Write an application to read temperature value in digital format and convert in to degree centigrade format. Write an LCD application to write temperature value in to LCD.		
Phase 3	Integration				
			Write a self-diagnostic test application to integrate all the peripherals and test the drivers. Deliverable: application		
Phase 4	Prepare Framework using KGDB				
			Prepare ADC Initialization & Operation Framework Deliverable: framework diagrams		